A Multimodal Approach for Detecting EEG Biomarkers During Acute Physical Activity

by

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Abstract

Performance biomarkers, traumatic brain injuries, and gait disorders are three areas of interest relevant to the field of sporting research. Electroencephalography (EEG) sensing offers a promising research avenue in each of these areas, but obstacles include immobile EEG devices and overwhelming movement artifacts. I propose a multimodal approach, by using a prototype EEG-based Mobile Brain and Body Imaging (MoBI) system to improve artifact removal and collect additional useful data modalities. The two aims for this project are (1) to validate the EEG-based MoBI system in an environment with significant movement artifacts, and (2) to examine the relationship between exercise intensity and specific EEG biomarkers. This project includes a significant device development and validation stage, followed by experimental data collection using a dual-task treadmill and oddball test protocol. In the proposed thesis, I expect to test three biomarker hypotheses and advance an EEG-based MoBI approach to sports research.

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1: Introduction and Motivation

Recent improvements in portable electroencephalography (EEG) devices represent a significant opportunity to examine cognitive performance [1], [2] and the effects of head impacts [3] during acute physical activity. However, artifacts from muscle activation, head motion, and electrode displacements are still major obstacles for the application of EEG to research in various sporting environments [4]. This remains true despite recent software developments in hybrid motion-artifact removal algorithms [5]. Therefore, there is a need to integrate additional sensor modalities like accelerometers and electrocardiography (ECG) electrodes to help eliminate artifacts and extract research-grade EEG data.

Portable EEG devices also represent an opportunity for Mobile Brain and Body Imaging (MoBI), which is an emerging research approach characterized by multimodal data collection outside a stationary laboratory environment. Along with artifact removal, additional data modalities can provide valuable data for researchers and clinicians. For example, EMG data can be used in combination with EEG to show the strength of EEG-EMG coherence during activity [6]. EEG-EMG coherence, pending further research, is becoming a valuable metric for studying gait disorders that lead to falling and head injuries [6]. In summary, an EEG-focused MoBI system that can provide research-grade data during different levels of physical activity is desirable.

2: Literature Review

2.1 EEG Context

EEG has been used in research since 1929 to record electrical activity from the human brain, usually using non-invasive electrodes on the scalp to measure small potentials generated by cortical neurons [7]. These electrodes are placed according to standard layouts (e.g. the 10-20 system) and are referenced against an electrode on a non-active body part, such as an earlobe. Data is displayed in time domain form or in frequency domain form (otherwise known as a spectral analysis) [8]. In time-domain form, multiple EEG recordings in response to a certain stimulus can be averaged to show event-related potentials (ERPs) [7]. In frequency-domain form, EEG data is divided into frequency bands like Delta (<4Hz), Theta (4-7 Hz), Mu (8-12 Hz), Alpha (8-15 Hz), and Beta (16-31Hz) depending on frequency and location on the scalp [8]. These are the most common analysis tools used in traditional EEG research.

Unfortunately, EEG research has also traditionally been largely unimodal and immobile due to various technological limitations. As a result, well-studied areas in the physical exercise domain include laboratory EEG tests of athletes compared to inactive people, and neurological after-effects of acute exercise [4]. Due to motion artifacts and a lack of portability, traditional EEG isn't suitable to show acute cognitive changes during exercise [5]. In recent years, improvements in portable EEG headsets and the Mobile Brain/Body Imaging (MoBI) approach have led to an increase in the collection of EEG data during exercise [4]. However, artifacts remain a major obstacle [5].

2.2 Artifacts and EEG-Based MoBI

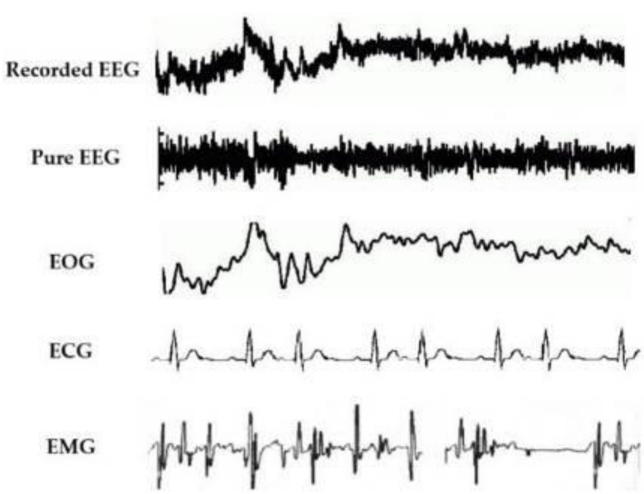

Figure 1: Raw EEG data composed of the EEG signal and EOG, ECG, and EMG artifacts [9]

Artifacts refer to any signal other than the desired scalp-EEG signal. With surface amplitudes of 0.001-0.01 mV, scalp EEG signals are often overwhelmed by surface electromyographic (EMG), electrooculographic (EOG), and electrocardiographic (ECG) voltages that have amplitudes of up to 10 mV, 0.1 mV, and 5 mV, respectively [7]. Figure 1 shows raw EEG data broken into the pure EEG signal with the artifact components from EOG, ECG, and EMG [9]. Artifacts from head motions and electrode displacements also pose significant challenges [10]. All these phenomena are emphasized during physical activity, so to maximize effectiveness of EEG artifact-removal, a combination of both software techniques and hardware techniques are required.

There are a wide variety of software techniques to remove artifacts from EEG data, with no consensus on an optimal solution for all artifacts [9]. Four widely used methods are Regression, Blind-Source Separation (BSS) including independent component analysis and canonical correlation analysis (ICA/CCA), Wavelet Transform, and Empirical-mode Decomposition (EMD); most current researchers use hybridized approaches [11]. Regression is a traditional method that is considered the gold-standard of artifact removal, but it requires at least one artifact-free reference electrode which makes it unsuitable for portable EEG systems [9]. CCA or ICA don't require artifact-free reference electrodes, but they can be ineffective with low-channel electrode setups [11]. Therefore, hybrid approaches such as EMD-ICA or Wavelet-ICA are often used in state-of-the-art artifact removal algorithms for walking and running [12]. Even these approaches, however, are often insufficient to remove all motion artifacts without over-cleaning the EEG data [10].

Therefore, incorporating hardware techniques to help with artifact removal is often necessary. A simple method used in recent literature is to add a time-synchronized head accelerometer as a reference node for walking frequency [12]. This allows gait frequency movement artifacts to be easily removed by assuming that the accelerometer displays purely artifact data [12]. The accelerometer also potentially helps with head-movement artifacts. Similar methods are used to remove EOG artifacts, using electrodes placed near the eye, and ECG artifacts, using a wearable ECG electrode [10]. This is generally done by displaying the frequencies of the ECG or EOG data alone, and removing it from the frequency-domain EEG data [10]. Unfortunately, EMG artifacts cannot be easily removed in a similar manner because they originate from many locations, so software techniques are required.

2.3 Applications of EEG-Based MoBI

One application of an EEG-based MoBI system is the quantification of head impacts outside a laboratory. Physical trauma to the head that causes shear or compression forces on the brain can lead to traumatic brain injuries (TBIs) or mild TBIs (mTBIs) [3]. TBIs contribute to more death and disability than any other traumatic insult [13]. In recent years, TBI researchers have shown increased interest in using EEG since spectral analysis shows promise as an mTBI detection tool [3]. However, acute neurological changes from real-world head impacts are relatively under-examined due to technological limitations. Some studies using spectral analysis have shown acute decreases in Alpha power and relative increases in Delta, Beta, and Theta power in injured groups [12], but according to a 2015 review, more research is required [3]. This application indicates that relative power spectral analysis is a necessary function for any research-grade MoBI system.

Another critical application of an EEG-based MoBI system is the field of neural biomarkers for sport performance [4]. 'Neural biomarkers of performance' refer to EEG signals that indicate factors like motor control, selective attention, and working memory which are crucial to high performance in sport [4]. One of the main biomarkers studied are ERPs like the P300. The P300 is a positive voltage pulse in time-domain that occurs after approximately 300ms after stimulus. Usually it is detected using an auditory or visual oddball paradigm, which is a test consisting of a series of repetitive stimuli interrupted by a deviant stimulus that elicits the P300 [15]. It is best detected by parietal-occipital electrodes, or Fz, Cz, Pz, and Oz [16], [17]. P300 amplitude is related to resource allocation and P300 latency is related to processing speed [15], which have been extensively shown to increase and decrease respectively in response to acute exercise [18]. However, a 2016 review found only 4 studies that acquired ERP data during exercise [4], indicating a clear need for research-grade MoBI systems in this area.

A final application of an EEG-based MoBI system is in the study of EEG-EMG coherence and gait. Each year, approximately one out of three adults above the age of 65 fall [6], usually as a result of gait disorders [19]. Falls are the leading cause of spine fractures, hip fractures, and TBI in older adults [6]. 'EEG-EMG Coherence' refers to the coupling of motor-cortex EEG signals with EMG signals from muscle activity, demonstrating a two-way cortico-muscular relationship [20]. It is best-detected by electrodes over the motor cortex (C3–F3, Cz–Fz and C4–F4) [21], and is calculated as the square modulus of the normalized cross-power spectral density for input signal x and output signal y at frequency λ, according to [22]

$$\left|C_{xy}(\lambda)\right|^2 = \left|\frac{f_{xy}(\lambda)}{\sqrt{f_{xx}(\lambda)f_{yy}(\lambda)}}\right|^2$$

This EEG-EMG coherence metric has shown promise in recent literature [19], [20], [22] for the study and treatment of gait disorders. However, improvement in artifact-removal is required before coherence can be studied beyond uncomplicated treadmill walking [20]. Therefore, there is need for a research-grade MoBI system that can demonstrate coherence during exercise that is more intense than a slow walk.

3: Approach

The first aim for this project is to develop and validate the effectiveness of a portable EEG, EMG, ECG, and accelerometer system for research. This means that device development and validation will be a significant portion of the project before data collection is possible. With this in mind, Figure 2 shows an approximate timeline for this project, with key deadlines (including course-related deadlines) highlighted:

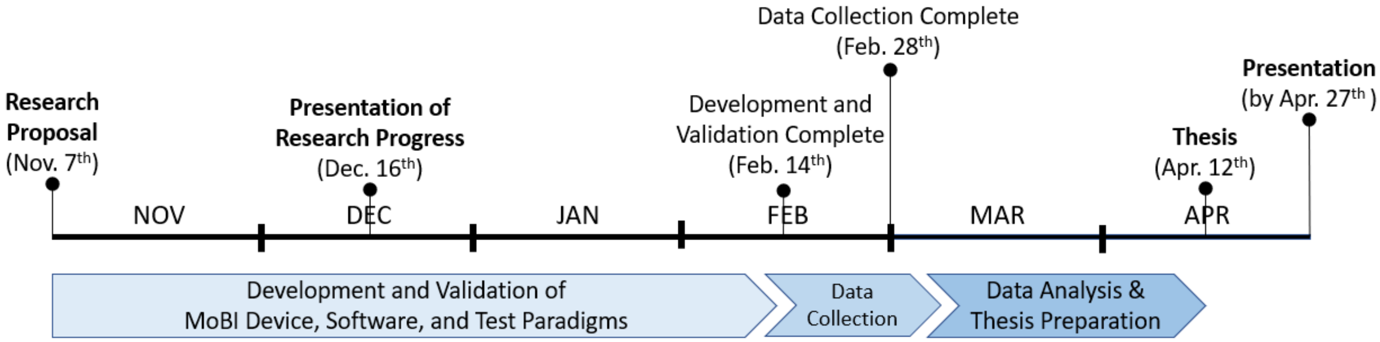

Figure 2: Project timeline showing course-related deadlines in bold.

The second aim of my thesis is to examine how different levels of physical activity (slow walking, quick walking, slow jogging, and quick jogging) affect:

a) Relative spectral band powers,
b) P300 amplitude and latency, and
c) EEG-EMG coherence.

Research during physical exercise for all the above metrics is sparse, as outlined in the *Literature Review*. However, based on adjacent findings, I can hypothesize an expected outcome for each metric respectively:

a) With increasing exercise intensity, relative alpha spectral power seems likely to decrease. This is because increased cognitive load which has been linked to a decrease in alpha power in stationary studies [8], and increased exercise intensity should mean increased cognitive load.
b) With increasing exercise intensity, P300 amplitude and latency seems likely to decrease and decrease, respectively. This is because P300 amplitude is related to the cognitive energy dedicated to a test paradigm [15], which will reduce as the cognitive load of movement increases. P300 latency has been shown to decrease immediately after exercise [4], which may hold true during exercise too.
c) With increasing exercise intensity, EEG-EMG coherence seems likely to get stronger. This is because coherence increases with increased muscle activation [22], and higher levels of intensity require more intense muscle contractions.

The proposed thesis will attempt to validate or invalidate these hypotheses.

4: Methodology

4.1 Equipment Development

The main piece of equipment required for this project is the EEG-based MoBI system, which requires significant development and validation. This system will incorporate EEG, EMG, ECG, and accelerometer data modalities, as shown in Figure 3.

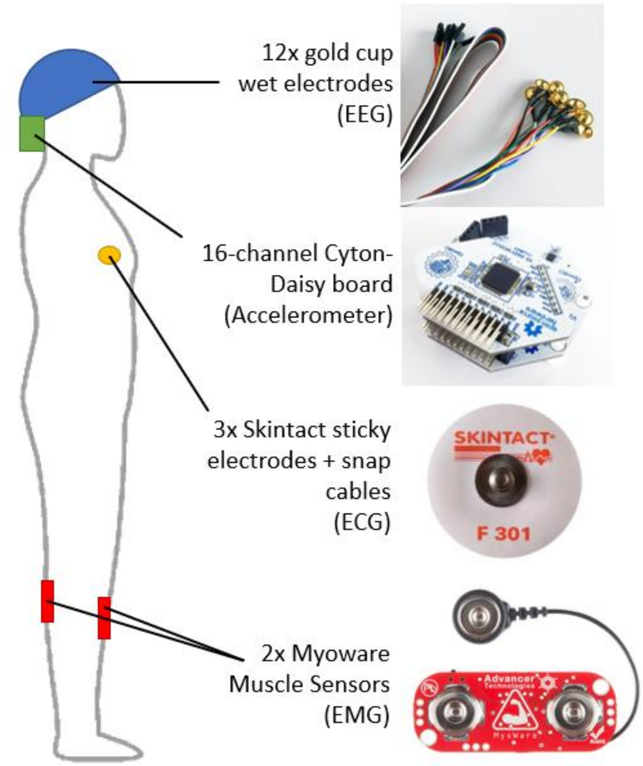

Figure 3: Sensors used for the EEG-based MoBI system

First, all data modalities will be collected using a Cyton-Daisy Biosensing board which has 16 channels and can connect using Bluetooth. The benefit of using this board for all data collection is the automatic time synchronization done by the OpenBCI software. Time synchronization is especially critical for EEG-EMG coherence and P300 detection [15]. The main drawback is the limitation of 16 channels, which means that the 19 electrodes required for a full 10-20 montage is not possible. However, sparser electrode setups have been shown to have negligibly different accuracy levels in P300 detection compared to a 19-electrode montage [17]. Also, headsets with fewer electrodes are more feasible in a mobile sporting environment. Therefore, this is an acceptable limitation.

EEG will be measured using 12 gold-cup wet electrodes on the scalp and referenced against the earlobes. The electrodes will be arranged according to the 10-20 system and will include locations F3, Fz, F4, C3, Cz, C4, Pz, and Oz [16], [17], which are optimized for P300 and EEG-EMG coherence detection. One proposed montage is shown in Figure 4, but this will be subject to further validation.

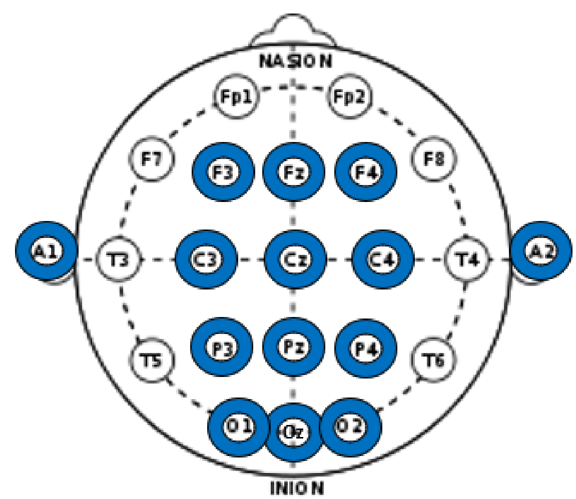

Figure 4: Electrode montage optimized for P300 and EEG-EMG coherence detection

EMG will be measured using 2 Myoware muscle sensors from OpenBCI on the anterior tibialis and lateral gastrocnemius muscles of the subject's dominant leg

[19]. These muscles produce significant surface EMG and are activated during standard gait. Shielded wiring fixed to the subject will need to connect the EMG sensors to the Cyton board near the head, which may cause artifacts during movement. If validation testing shows that this is a serious problem, wireless EMG sensors will be considered as discussed in *Appendix B: Alternative Equipment for EEG-based MoBI Device*.

ECG will be measured using 3 Skintact foam adhesive electrodes from OpenBCI, placed on the sternum and either side of the chest. They will be connected to the Cyton board using snap electrode cables from OpenBCI. Necessary amplification and rectification are performed on the Cyton board using OpenBCI software.

Accelerometer data will be measured using the Cyton on-board accelerometer. A shielded enclosure for the Cyton-Daisy board will need to be designed and printed using ABS plastic and foil. Additionally, a constraint system will need to be added to ensure the accelerometer is properly coupled with head movement, and not bouncing during the walking and running.
Other than the EEG-based MoBI device, this project requires a treadmill, screen, camera, and a computer running OpenBCI. A full list of required equipment is included in *Appendix A: List of Required Equipment*.

4.2 Equipment Validation

Validation of the MoBI system and relevant data analysis techniques will be necessary before formal data collection begins. Some validation tests for early in the development process, and their pass criteria, are listed in Table 1.

Table 1: Validation Tests for Development of the MoBI System

Validation Test	Pass Criterion
Detection of ocular artifacts (blinking), ECG signals, and muscle contractions while sitting and standing [7].	Qualitative confirmation to ensure the system functions.
Alpha wave power attenuation upon eye opening while sitting and standing.	Alpha band power during eyes-open divided by eyes-closed must be less than 1 [8].
P300 detection while sitting and standing using the visual oddball paradigm described in *Testing Protocol*	P300 ERPs must be qualitatively detected after a 5-minute visual oddball paradigm (30 deviant stimuli total).
EEG/EMG coherence detection using a seated button-push paradigm [21].	Coherence analysis must show significant coherence between EEG and EMG signals from the flexor digitorum superficialis muscle.
5-80 Hz Absolute EEG spectral power during walking and sitting should be approximately equal.	Total absolute spectral power during walking divided by sitting should be only slightly less than 1 [12].

Ultimately, the best form of validation will be performing the experimental test protocols on myself, with pass criteria being preliminary data analysis to extract the relevant biomarkers. This validation test will act as a final step before formal data collection begins.

4.3 Testing Protocol

The testing protocol summarized here is a synthesis of protocols used in acute exercise studies [1], [2] and dual-task gait studies [18], [19] along with some novel elements. It will be subject to changes based on preliminary validation steps.

The experimental setup is a treadmill with a screen mounted at approximately eye-level. The study participants, expected to be 5-8 lab students, will be given time to get used to walking/jogging on the treadmill while holding a small computer mouse to reduce likelihood of injury; treadmill speeds will also be relatively low. They will each be outfitted with the EEG-based MoBI device, with EMG electrodes on their dominant leg. A camera will film all data collection.

Subjects will be asked to perform a visual oddball paradigm task for 5 minutes at each activity level:

1. Seated (~0 km/hr)
2. Walking slowly (~2.5 km/hr)
3. Walking quickly (~5.0 km/hr)
4. Jogging slowly (~7.0 km/hr)
5. Jogging quickly (~8.0 km/hr) [18]

Treadmill speeds will be the same for all participants. The goal is not to exercise to exhaustion, so the highest speed (~8.0 km/hr) will have to be validated to ensure all participants can relatively easily jog for 5 minutes at that speed. Participants will also be instructed to keep gait as constant as possible throughout the task to help with artifact removal.

The visual oddball paradigm task will be displayed on the screen. Participants will be asked to click a mouse button upon seeing a specific image, the 'target' image, which will randomly appear 10% of the time [1]. Non-target unrelated images will appear the rest of the time, with images switching at a frequency of ~1 Hz [1].

4.4 Data Analysis

Data analysis will make use of the lab's existing MATLAB functions that were already validated for artifact removal from gait-related dataset. These include functions for ICA, CCA, EMD-CCA, Wavelet-CCA, ICA-EMD, and ICA-Wavelet (see *Artifacts and EEG-based MoBI* section for acronyms). They also include functions that use accelerometer data to help with artifact removal. Additional development work will be required to incorporate ECG data into the MATLAB artifact-removal functions in the same way that accelerometer data is used. Also, determining the most effective combination of filtering and functions will require validation: See the *Equipment Validation* section for potential tests that can be used to identify effective function algorithms. Ideally before experimental data collection, documentation of a standard artifact-removal algorithm for walking and jogging with the EEG-based MoBI system be completed.

5: Impact

To conclude, this project has four expected deliverables:

1. An EEG-based MoBI system that incorporates EEG, EMG, ECG, and accelerometer data modalities.
2. An EEG artifact-removal software methodology that incorporates ECG and accelerometer data.
3. A dataset from 5-8 participants on a treadmill at different speeds while performing a dual-task paradigm.
4. A thesis paper validating 1 and 2 for detection of P300 ERPs, power spectral analysis, and EEG-EMG coherence. It will also show how these biomarkers are affected by different intensities of walking and running.

After project completion, the EEG-based MoBI system and artifact-removal algorithm will continue to be applicable to a variety of fields of research. It will be a valuable tool for studying TBI in more realistic scenarios, for example. It should also advance research into sport-related performance biomarkers by enabling data collection despite significant artifact contaminations. Gait disorders, too, represent a research field that needs a portable and robust MoBI system like the one developed for this project. The EEG-based MoBI system could also be further improved upon with proprietary dry electrodes, PCB boards, and/or headcap components. See *Appendix B: Alternative Equipment for EEG-based MoBI Device* for more

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The dataset collected during the study will also be valuable. Additional potential analyses could include looking for event-related synchronization/desynchronization (ERD/ERS) and Movement-related Cortical Potentials (MRCP) using the EMG activations as a trigger. You could also compare this dataset to one from older participants or stroke patients to find how P300, spectral analysis, and EEG-EMG coherence differs among various groups.

Lastly, it's my hope that the thesis may evolve into a strong contribution to the research fields of EEG, MoBI, and physical exercise. The *Literature Review* section of this proposal includes multiple experimental studies from the past 5 years with adjacent hypotheses in those fields [1], [2], [4], [6], [12], [19], [22]. I believe that this thesis fills a valuable gap in the research that those studies have not yet covered.

6: References

[1] T. Bullock, H. Cecotti, and B. Giesbrecht, "Multiple stages of information processing are modulated during acute bouts of exercise," *Neuroscience*, vol. 307, pp. 138–150, Oct. 2015, doi: 10.1016/j.neuroscience.2015.08.046.

[2] S. Ludyga, T. Gronwald, and K. Hottenrott, "Effects of high vs. low cadence training on cyclists' brain cortical activity during exercise," *Journal of Science and Medicine in Sport*, vol. 19, no. 4, pp. 342–347, Apr. 2016, doi: 10.1016/j.jsams.2015.04.003.

[3] P. E. Rapp *et al.*, "Traumatic Brain Injury Detection Using Electrophysiological Methods," *Front. Hum. Neurosci.*, vol. 9, 2015, doi: 10.3389/fnhum.2015.00011.

[4] G. Cheron *et al.*, "Brain Oscillations in Sport: Toward EEG Biomarkers of Performance," *Front. Psychol.*, vol. 7, 2016, doi: 10.3389/fpsyg.2016.00246.

[5] K. Gramann, D. P. Ferris, J. Gwin, and S. Makeig, "Imaging natural cognition in action," *International Journal of Psychophysiology*, vol. 91, no. 1, pp. 22–29, Jan. 2014, doi: 10.1016/j.ijpsycho.2013.09.003.

[6] D. De Venuto, V. Annese, M. Ruta, and E. Di Sciascio, "Wireless Control of the Cortico-Muscolar Coupling in Human Walking to Prevent Falls," Dec. 2015.

[7] E. Niedermeyer, "The Clinical Relevance of EEG Interpretation," *Clinical Electroencephalography*, vol. 34, no. 3, pp. 93–98, Jul. 2003, doi: 10.1177/155005940303400303.

[8] O. M. Bazanova and D. Vernon, "Interpreting EEG alpha activity," *Neuroscience & Biobehavioral Reviews*, vol. 44, pp. 94–110, Jul. 2014, doi: 10.1016/j.neubiorev.2013.05.007.

[9] X. Jiang, G.-B. Bian, and Z. Tian, "Removal of Artifacts from EEG Signals: A Review," *Sensors (Basel)*, vol. 19, no. 5, Feb. 2019, doi: 10.3390/s19050987.

[10] M. K. Islam, A. Rastegarnia, and Z. Yang, "Methods for artifact detection and removal from scalp EEG: A review," *Neurophysiologie Clinique/Clinical Neurophysiology*, vol. 46, no. 4, pp. 287–305, Nov. 2016, doi: 10.1016/j.neucli.2016.07.002.

[11] K. L. Snyder, J. E. Kline, H. J. Huang, and D. P. Ferris, "Independent Component Analysis of Gait-Related Movement Artifact Recorded using EEG Electrodes during Treadmill Walking," *Front. Hum. Neurosci.*, vol. 9, 2015, doi: 10.3389/fnhum.2015.00639.

[12] E. Arad, R. P. Bartsch, J. W. Kantelhardt, and M. Plotnik, "Performance-based approach for movement artifact removal from electroencephalographic data recorded during locomotion," *PLoS One*, vol. 13, no. 5, May 2018, doi: 10.1371/journal.pone.0197153.

[13] M. C. Dewan *et al.*, "Estimating the global incidence of traumatic brain injury," *Journal of Neurosurgery*, vol. 130, no. 4, pp. 1080–1097, Apr. 2018, doi: 10.3171/2017.10.JNS17352.

[14] J. N. Ianof and R. Anghinah, "Traumatic brain injury: An EEG point of view," *Dement Neuropsychol*, vol. 11, no. 1, pp. 3–5, 2017, doi: 10.1590/1980-57642016dn11-010002.

[15] M. Hansenne, “Event-Related Brain Potentials in Psychopathology: Clinical and cognitive perspectives,” *Psychologica Belgica*, vol. 46, no. 1–2, Art. no. 1–2, Mar. 2006, doi: 10.5334/pb-46-1-2-5.

[16] J. Polich and J. R. Criado, “Neuropsychology and neuropharmacology of P3a and P3b,” *International Journal of Psychophysiology*, vol. 60, no. 2, pp. 172–185, May 2006, doi: 10.1016/j.ijpsycho.2005.12.012.

[17] M. T. McCann, D. E. Thompson, Z. H. Syed, and J. E. Huggins, “Electrode subset selection methods for an EEG-based P300 brain-computer interface,” *Disabil Rehabil Assist Technol*, vol. 10, no. 3, pp. 216–220, May 2015, doi: 10.3109/17483107.2014.884174.

[18] P. De Sanctis, J. S. Butler, B. R. Malcolm, and J. J. Foxe, “Recalibration of inhibitory control systems during walking-related dual-task interference: A Mobile Brain-Body Imaging (MOBI) Study,” *NeuroImage*, vol. 94, pp. 55–64, Jul. 2014, doi: 10.1016/j.neuroimage.2014.03.016.

[19] M. De Tommaso, E. Vecchio, K. Ricci, A. Montemurno, D. De Venuto, and V. F. Annese, “Combined EEG/EMG evaluation during a novel dual task paradigm for gait analysis,” in *Proceedings - 2015 6th IEEE International Workshop on Advances in Sensors and Interfaces, IWASI 2015*, Aug. 2015, pp. 181–186, doi: 10.1109/IWASI.2015.7184949.

[20] T. H. Petersen, M. Willerslev-Olsen, B. A. Conway, and J. B. Nielsen, “The motor cortex drives the muscles during walking in human subjects,” *J Physiol*, vol. 590, no. Pt 10, pp. 2443–2452, May 2012, doi: 10.1113/jphysiol.2012.227397.

[21] P. Brown, S. F. Farmer, D. M. Halliday, J. Marsden, and J. R. Rosenberg, “Coherent cortical and muscle discharge in cortical myoclonus,” *Brain*, vol. 122, no. 3, pp. 461–472, Mar. 1999, doi: 10.1093/brain/122.3.461.

[22] L. Roeder, T. W. Boonstra, S. S. Smith, and G. K. Kerr, “Dynamics of corticospinal motor control during overground and treadmill walking in humans,” *Journal of Neurophysiology*, vol. 120, no. 3, pp. 1017–1031, May 2018, doi: 10.1152/jn.00613.2017.

Appendix A: List of Required Equipment

Table A1 is meant as an exhaustive list of equipment required for the proposed project. However, changes and omissions are likely as the project requirements shift in the next few months.

Table A2: Equipment List

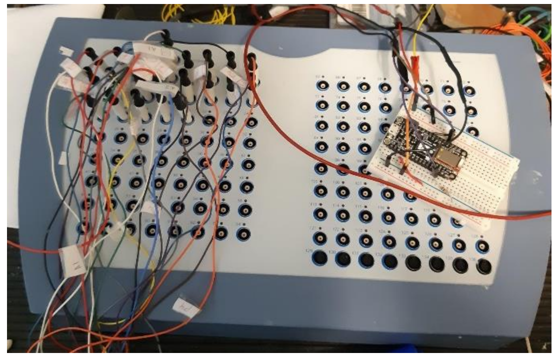
Equipment	Purpose	Location and/or Cost (USD)
Treadmill	For levels of physical activity	May have access to a personal one or might be one in a lab nearby. Otherwise ~800$
Mouse	Clicking during oddball paradigm	Own
Screen	Display visual oddball paradigm	Own
Computer running OpenBCI	Data collection	Own
Cyton board + Dongle + Battery pack	Data collection and portable transmission	Own
Daisy add-on	Adds 8 channels to Cyton Board	500$ from OpenBCI
Camera	For filming data collection	Own
EEG headset (3D printed fabric + electrode mounts)	Mounting wet electrodes	Own personal one, but may need to make more for test participants
Gold Electrodes	For EEG recording	Own
Wet electrode gel	For EEG recording	Own
Skin Prep gel	For EEG recording	Own
Snap electrodes	For ECG recording	40$ from OpenBCI
SkinTact Sticky electrodes	For ECG and EMG recording	13$ from OpenBCI, 30 pack
2x MyoWare Muscle Sensor	For EMG recording	80$ (40$ each) from OpenBCI
Shielded cabling	To connect EMG sensors with Cyton board	Available in the lab?
3D printing ABS plastic	For Cyton + Daisy housing	Own

Total cost is ~$633 USD (~$830 CAD) excluding treadmill, ~$1,433 USD ($1,850 CAD) including treadmill.

Appendix B: Alternative Equipment for EEG-based MoBI Device

Due to the significant time limitations on this project, I have opted for sensors that can become operational quickly in the *Equipment Development* section. These sensors have several drawbacks, such as the limited number of electrode channels. Here are some alternative sensors that could be considered for the future, or if problems arise with the chosen sensors:

1. **ANT EEG sensor** in the ICICS X015 lab: reads from 128 EEG electrodes simultaneously. Time-synchronization may become a problem.



2. **Dry electrodes** from SimPL or OpenBCI: reduce setup time but are more susceptible to electrode displacement artifacts. Currently in development at SimPL.

3. **Wireless EMG sensors:** if cable motion artifacts are a serious issue, may need to consider wireless EMG options

4. SimPL proprietary **PCB**: PCB currently being developed at SimPL that is smaller and lighter than the Cyton-Daisy board, and potential manages more channels.